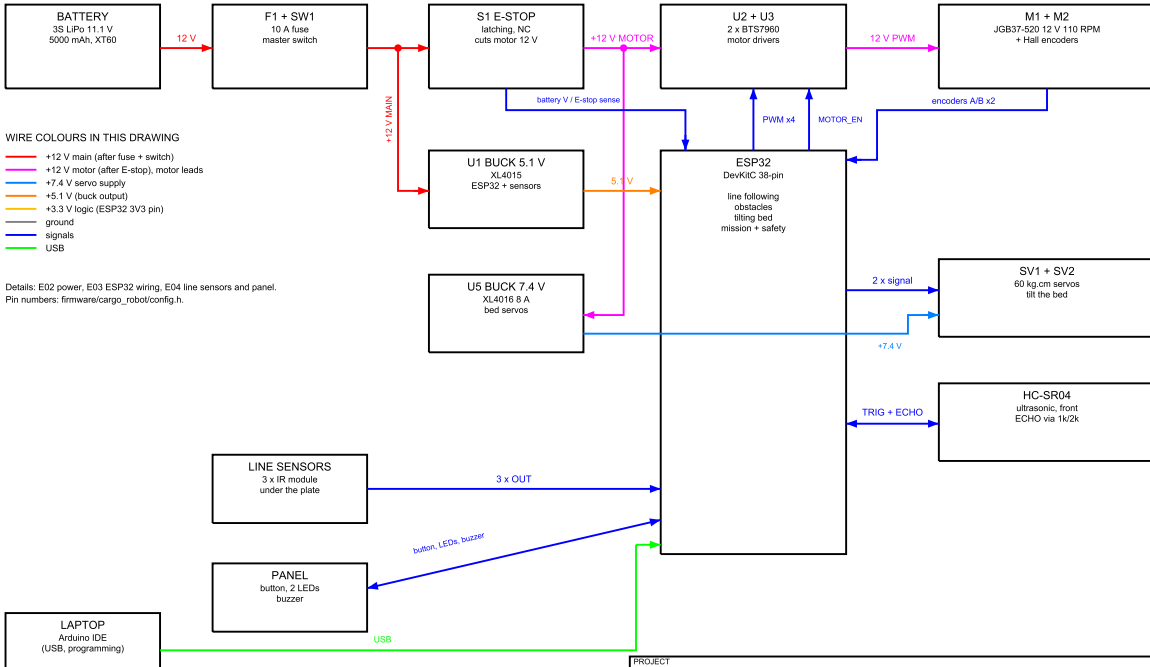
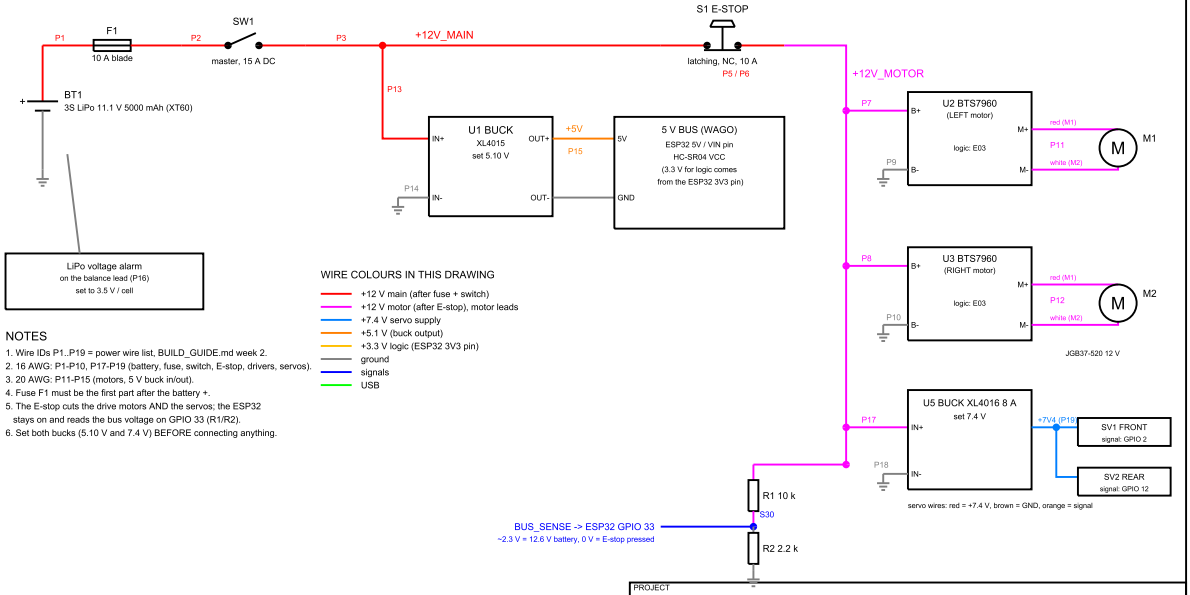


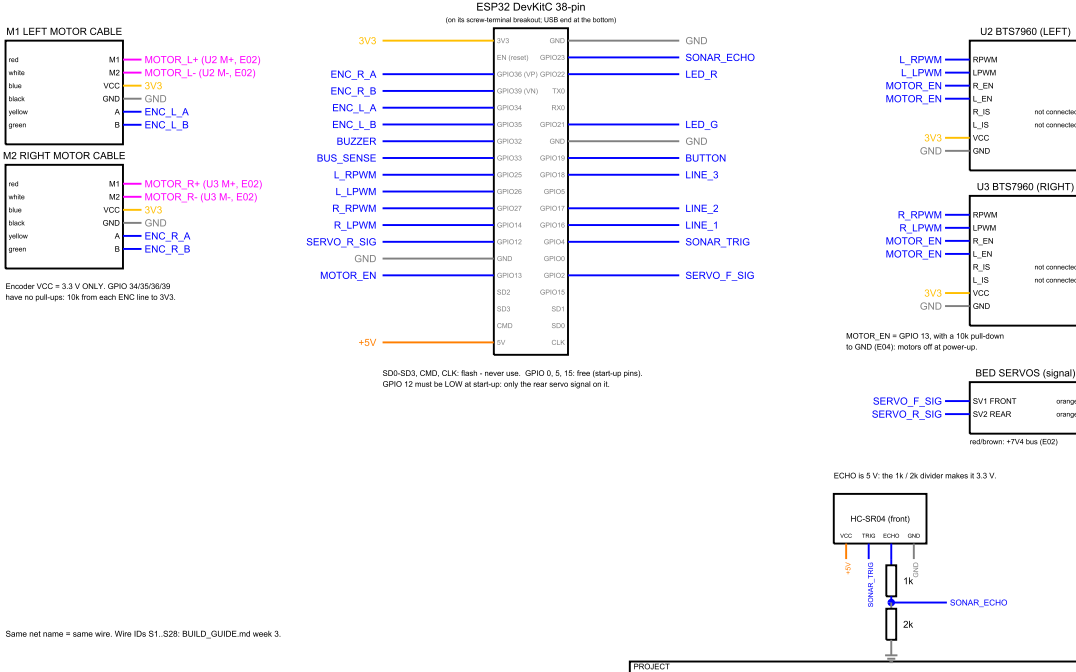
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: SYSTEM BLOCK DIAGRAM					DWG No.: CR-V2-E01
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 1 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 10/11/2026	DRAWN:



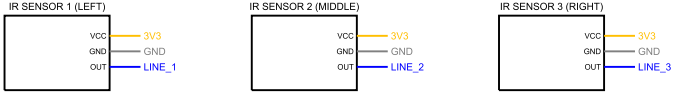
NOTES

1. Wire IDs P1..P19 = power wire list, BUILD_GUIDE.md week 2.
2. 16 AWG: P1-P10, P17-P19 (battery, fuse, switch, E-stop, drivers, servos).
3. 20 AWG: P11-P15 (motors, 5 V buck in/out).
4. Fuse F1 must be the first part after the battery +.
5. The E-stop cuts the drive motors AND the servos; the ESP32 stays on and reads the bus voltage on GPIO 33 (R1/R2).
6. Set both bucks (5.10 V and 7.4 V) BEFORE connecting anything.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: POWER DISTRIBUTION					DWG No.: CR-V2-E02
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 2 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 10/11/2026	DRAWN:

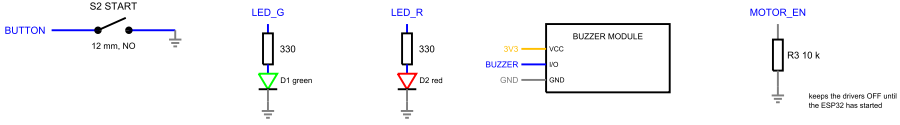


PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: ESP32 WIRING					DWG No.: CR-V2-E03
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 3 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 10/11/2026	DRAWN:



3-pin IR obstacle-avoidance modules (FC-51 type), on the printed bar under the bottom plate. LINE_1 = GPIO 16, LINE_2 = GPIO 17, LINE_3 = GPIO 18.
Power them from 3V3, NOT 5 V: then their OUT is safe for the ESP32. Each blue pot: set so the module's LED switches between the floor and the black tape.

PANEL (BUTTON = GPIO 19, LED_G = 21, LED_R = 22, BUZZER = 32) AND PULL-DOWN (on the I/O perfboard)



NOTES

1. All logic on this sheet runs on 3.3 V from the ESP32's 3V3 pin. No I2C and no I/O expander.
2. Line sensors: output HIGH over black tape (setting line_black). Check with the "line" command.
3. Same net name = same wire. Pin numbers: firmware/cargo_robot/config.h.

WIRE COLOURS IN THIS DRAWING

- +12 V main (after fuse + switch)
- +12 V motor (after E-stop), motor leads
- +7.4 V servo supply
- +5.1 V (buck output)
- +3.3 V logic (ESP32 3V3 pin)
- ground
- signals
- USB

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LINE SENSORS AND PANEL					DWG No.: CR-V2-E04
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 4 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 10/11/2026	DRAWN: